

BHAVYA RUSTAGI

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WORK EXPERIENCE

Newgen Software Technologies Limited

June 2026 – September 2026

AI Engineer Intern

India (Remote)

- Built an **enterprise semantic search platform** enabling natural-language retrieval across **PDF, DOCX**, and **TEXT** documents using dense vector embeddings and semantic similarity search.
- Engineered document ingestion, text extraction, semantic chunking, embedding generation, and **MongoDB Atlas Vector Search** pipelines for context-aware retrieval with source attribution.
- Implemented **OCR** asynchronous document processing, grounded AI answer generation, citation validation, and resilient fallback retrieval for reliable enterprise document search.

AirDawg Labs

April 2026 – May 2026

AI Systems Evaluation Intern

India (Remote)

- Designed **Docker-based** CLI benchmark tasks to evaluate **AI coding agents** across real world software engineering workflows, ensuring reliable execution, reproducible testing, and consistent benchmark results.
- Built **reference solutions**, automated test suites, and evaluation pipelines using **Docker**, **Cursor**, and AI assisted development workflows to validate correctness, scalability, and benchmark consistency.
- Improved benchmark reliability through structured testing, prompt engineering, and failure analysis of **LLM agents**, identifying edge cases, improving evaluation quality, and reducing inconsistent agent behavior.

PROJECTS

GitHub Issue → PR Autonomous Coding Agent — *Next.js, FastAPI, PostgreSQL, Redis*, [GitHub](#) — [Live](#)
Docker, OpenAI

- Built an autonomous engineering agent that transforms bounded **GitHub issues into pull requests** through repository analysis, structured planning, sandboxed implementation, verification, repair, and human approval.
- Engineered durable orchestration using **FastAPI, PostgreSQL, Redis, Dramatiq, SSE**, and **Docker**, supporting retries, cancellation, immutable repository context, GitHub App authorization, and secure execution.

AgentScope — *Next.js, FastAPI, PostgreSQL, Docker, OpenTelemetry*

[GitHub](#) — [Live](#)

- Built an **agent evaluation and observability platform** for tracing AI-agent executions, scoring outputs with rubric/LLM-based evaluators, monitoring quality drift, and detecting regressions across agent versions.
- Engineered **experiment comparison, statistical calibration, automated regression detection and git bisection, live monitoring/incident workflows, and persistent execution tracing** using FastAPI, PostgreSQL, Docker, and OpenTelemetry-style instrumentation.

TECHNICAL SKILLS

Languages	Python, TypeScript, SQL
AI/LLM	RAG, LLM Agents, Embeddings, Semantic Search, MCP, Tool Calling , Prompt Engineering
Agent Engineering	Agent Orchestration, Human-in-the-Loop, Guardrails, Structured Outputs
Frameworks	FastAPI, React, Next.js, Node.js
Databases	PostgreSQL, MongoDB Atlas, Redis, MongoDB Atlas Vector Search
Observability	LLM-as-a-Judge, Rubric Evaluation, Statistical Calibration, Regression Detection
Backend Engineering	REST APIs, Async Processing, Background Jobs, SSE, Idempotency, Retry/Backoff
Tools	Docker, Git, GitHub, GitHub App/API, Dramatiq, Pytest, OpenAI API

EDUCATION

Maharaja Surajmal Institute of Technology (MSIT)

2023 – Present

B.Tech in Information Technology, CGPA: 7.94

Delhi, India